

Tensor Network techniques for differential equations

QTADE School, Bilbao

Raghavendra Dheeraj Peddinti

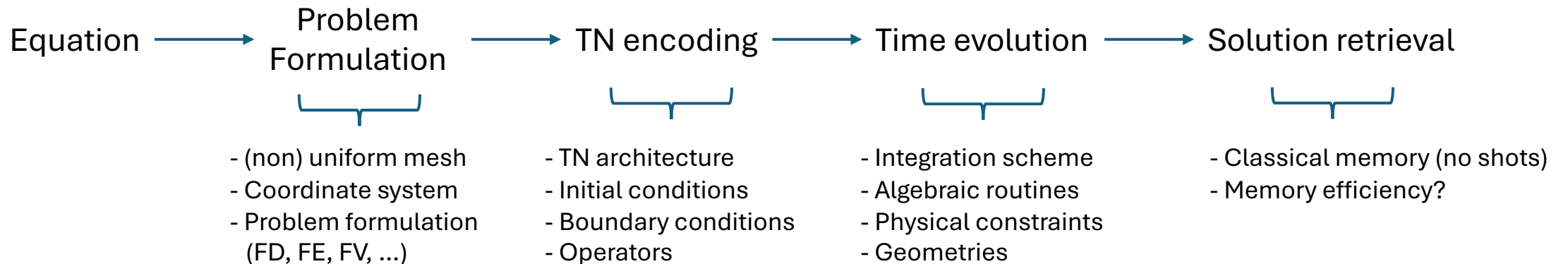
Stefano Pisoni

Quantum Research Center, TII, Abu Dhabi

TUHH, Hamburg

What the lectures are about – high level

Use Tensor Networks (TNs) to exploit the low-rank structure of solutions and operators defining PDEs (no quantum)



Rank / Bond dimension: χ

Structure of the lectures

Today:

- Part I: Representing fields and operators (Stefano)
- Part II: Linear algebraic routines (Raghavendra)

Tomorrow afternoon:

... with Tensor Networks

- Part III: The forward problem: time evolution (Stefano)
- Part IV: The inverse problem: data-driven methods (Raghavendra)

Computational Fluid Dynamics with QTT

Why fluid dynamics

CFD framework

Numerical results

Recent literature

Toward turbulence

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Why fluid dynamics

After yesterday, we now have all the QTT primitives to solve PDEs
Which one should we solve?

The question is not trivial $\rightarrow$ QTT compress information in a peculiar way

Rank/BD = non factorizability $\rightarrow$ entanglement in quantum d.o.f. (area law $\rightarrow$ efficient MPS)

Rank in QTT (binary discretization of the domain) $\rightarrow$ interscale correlations

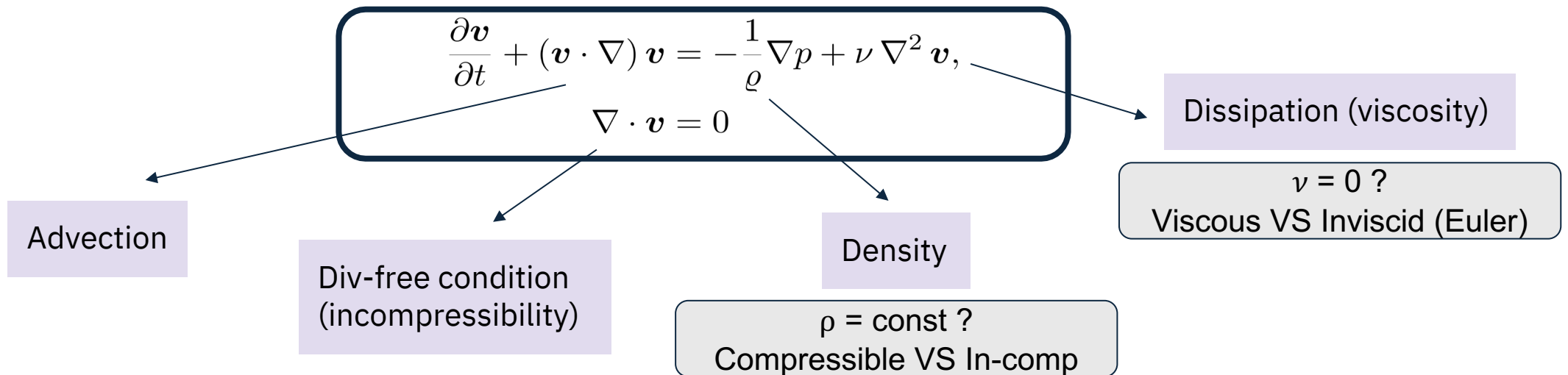
So what about fluid flows? We need to talk about turbulence (notion of scale invariance and scale locality)

Turbulent regimes are the computational bottleneck of CFD

Let's start by the governing equations!

Why fluid dynamics

The Navier-Stokes equations

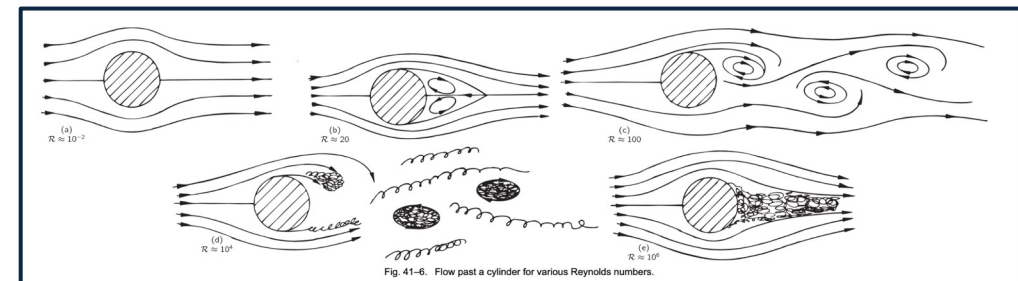


$$\text{Re} \sim \frac{\text{advection}}{\text{dissipation}} \sim \frac{UL}{\nu}$$

Laminar

Turbulent

(Re >> 1)

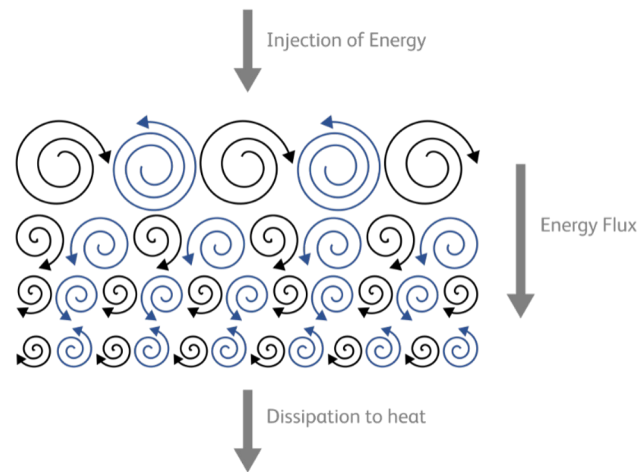


Why fluid dynamics

Turbulence

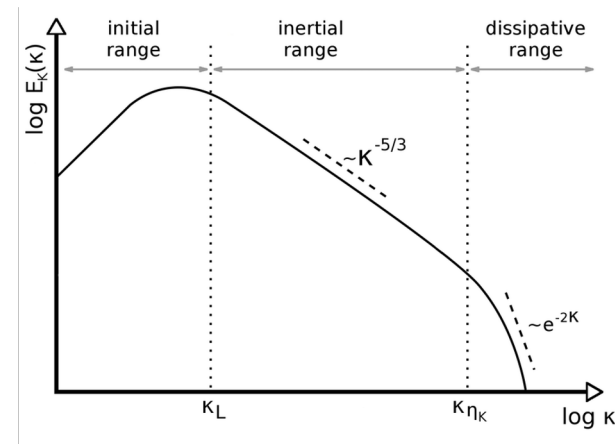
Richardson 1922

*"Big whorls have little whorls that feed on their velocity,
And little whorls have lesser whorls and so on to viscosity."*



Self similarity within the inertial range!

Energy cascade - Kolmogorov 1941



Stronger coupling between neighboring scales

$$L/\eta = \text{Re}^{3D/4}$$

Re	# scales
10E4	10
10E5	12/13
10E6	15
10E7	17/18
10E8	20
10E9	22/23

Why fluid dynamics

Direcet Numerical Simulations (DNS)

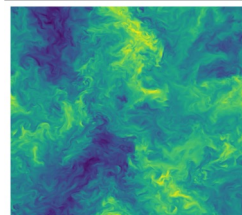
Direct Numerical Simulations (DNS)

Number of 'pixels' required to resolve all details:

$$\text{Re}^{9/4}$$

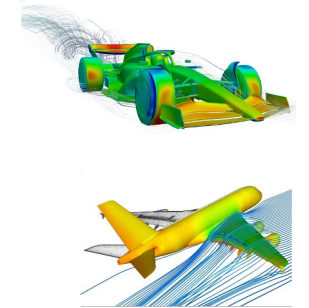
Typical airfoil simulation,
 $\text{Re} \sim 2\,000\,000 - 20\,000\,000$,
 $\Leftrightarrow$ number of 'pixels' $\sim 10^{13} - 10^{16}$

DNS dataset, $\text{Re} \sim 15000$
(Madrid, 2017)



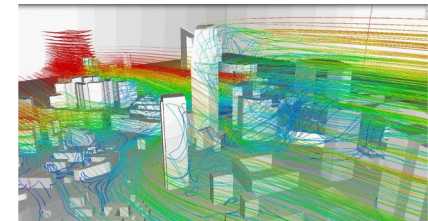
CFD scenario:

- Fluid flow predictions
- Shape optimization
- Forecasting



Hands-on expertise:

- Mesh generation
- Boundary conditions
- FD, FEM, ...
- Time evolution scheme
- Ad-hoc phenomenological models (RANS/LES...)



Why fluid dynamics

Turbulence

QTT tensors represent scales



Scale-locality/invariance in turbulence



Low-rank representation for turbulent fields?

If the correlations among scales are mostly “local”, then we expect a low rank structure
→ Logarithmic memory cost in the mesh size $\sim N\chi^2$

This idea was firstly laid down by Gourianov in 2022, results:

Why fluid dynamics

First paper on this topic

A quantum-inspired approach to exploit turbulence structures

N Gourianov, M Lubasch, S Dolgov, QY van den Berg, H Babaei, P Givi, ...
Nature Computational Science 2 (1), 30-37

190

2022

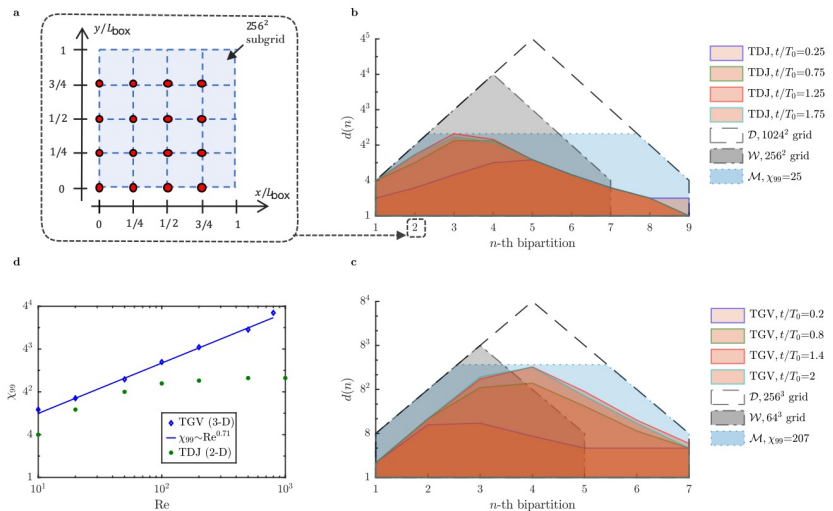


FIG. 1: Interscale correlations of turbulent fluid flows. (a) and (b) correspond to a square with edge length

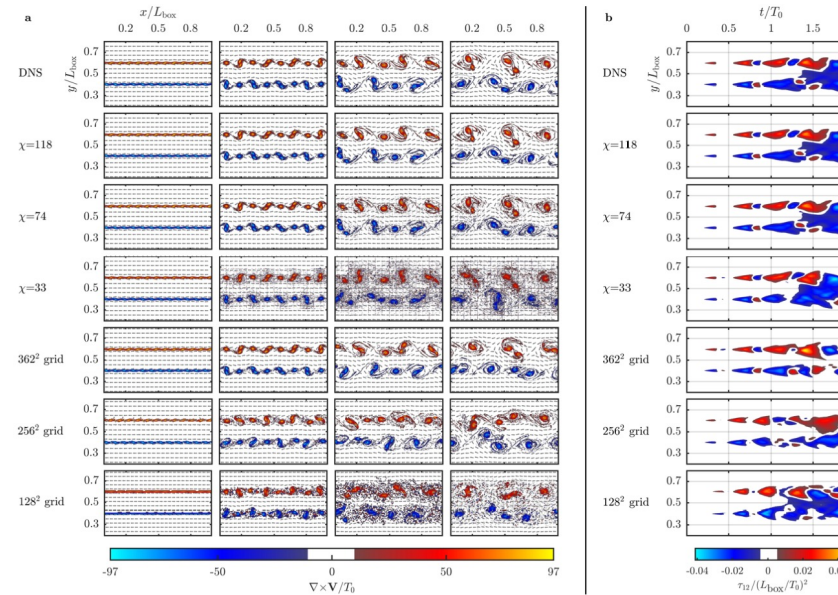


FIG. 2: 2-D Temporally developing jet. Dynamical simulation of the INSE in 2-D for a planar jet streaming along x with $Re = 1000$, as defined in the *Set-up of numerical experiments* section in *Methods*. (a) shows snapshots

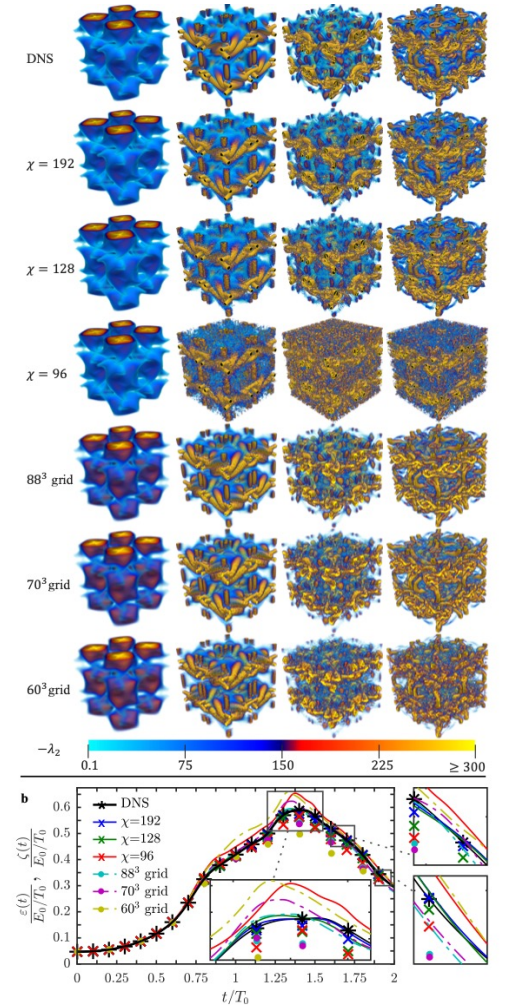


FIG. 3: 3-D Taylor-Green vortex. Dynamical simulation of the INSE in 3-D for the Taylor-Green vortex and $Re = 800$, as defined in the *Set-up of*

Computational Fluid Dynamics with QTT

Why fluid dynamics

CFD framework

Numerical results

Recent literature

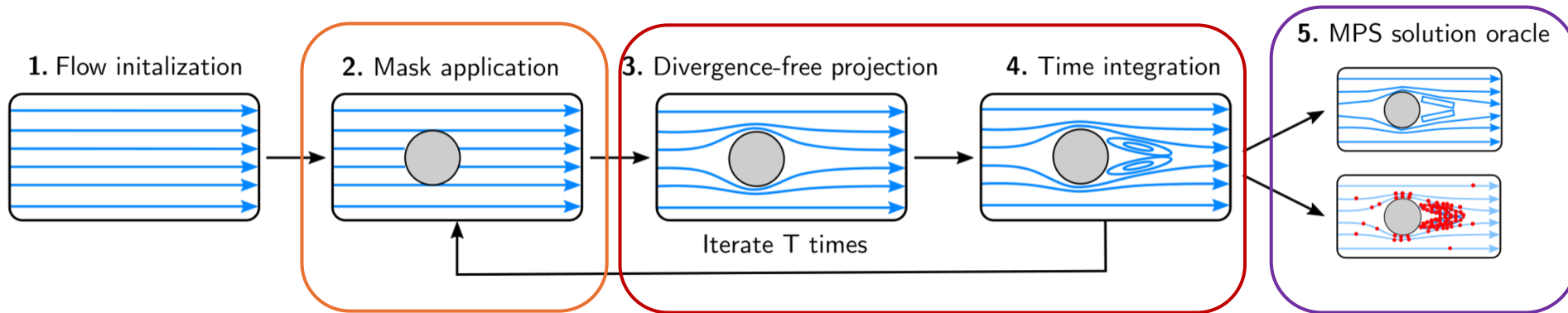
Toward turbulence

CFD framework

Problem: CFD pipelines are very versatile → can we replicate in QTT? (with better scaling)

(Very active research field, list of papers at the end of the section)

We follow Peddinti et al. (2023)



Approximate MPS masks of arbitrary objects

Time evolution w/ open boundary conditions and divergence-free projection

Efficient retrieval from MPS

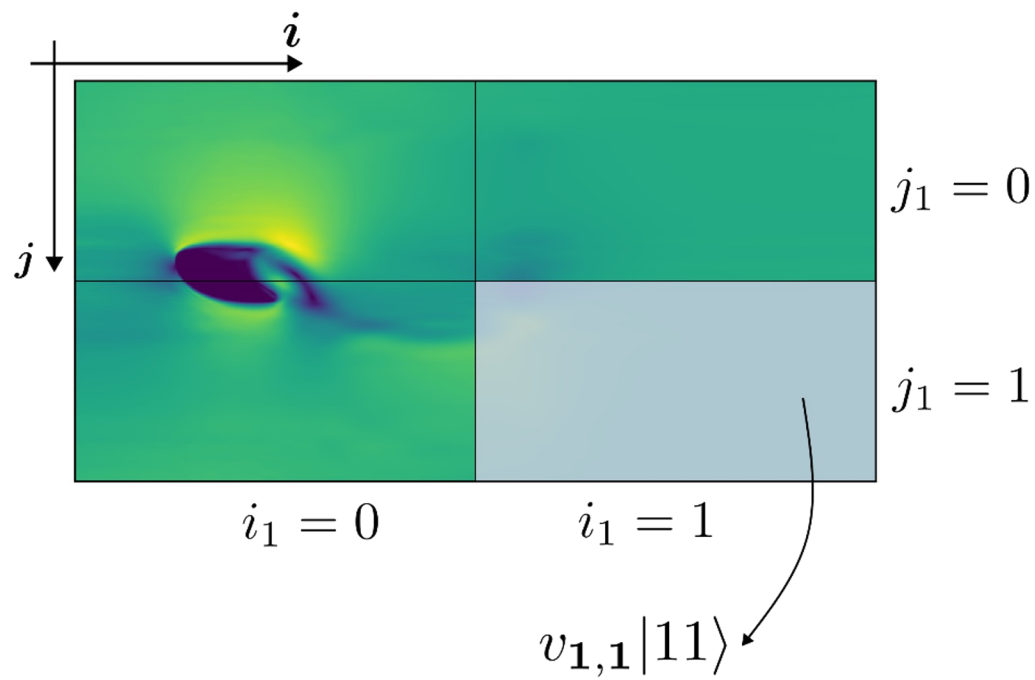
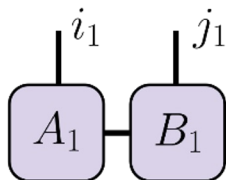
CFD framework

First step: Flow initialization
QTT encoding recap

2D stacked

$$v(x, y) \xrightarrow{\text{discretize}} v(\mathbf{i}, \mathbf{j}) \xrightarrow[\text{encoding}]{\text{amplitude}} v_{\mathbf{i}, \mathbf{j}} |\mathbf{i}, \mathbf{j}\rangle$$

$$v_{\mathbf{i}, \mathbf{j}} = A_1^{(i_1)} B_1^{(j_1)}$$



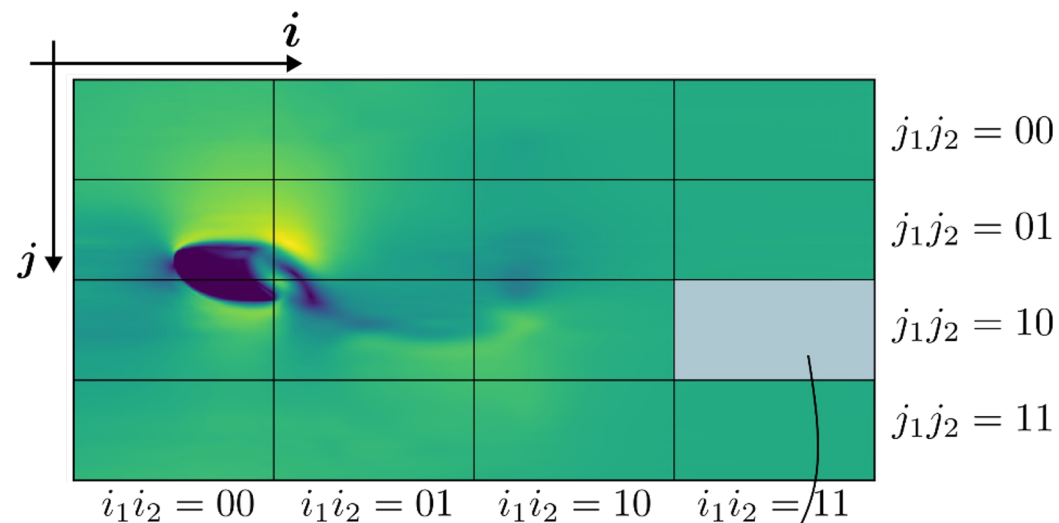
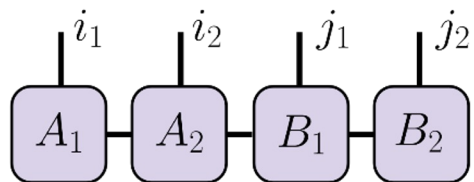
CFD framework

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$$v_{\mathbf{i}, \mathbf{j}} = A_1^{(i_1)} A_2^{(i_2)} B_1^{(j_1)} B_2^{(j_2)}$$



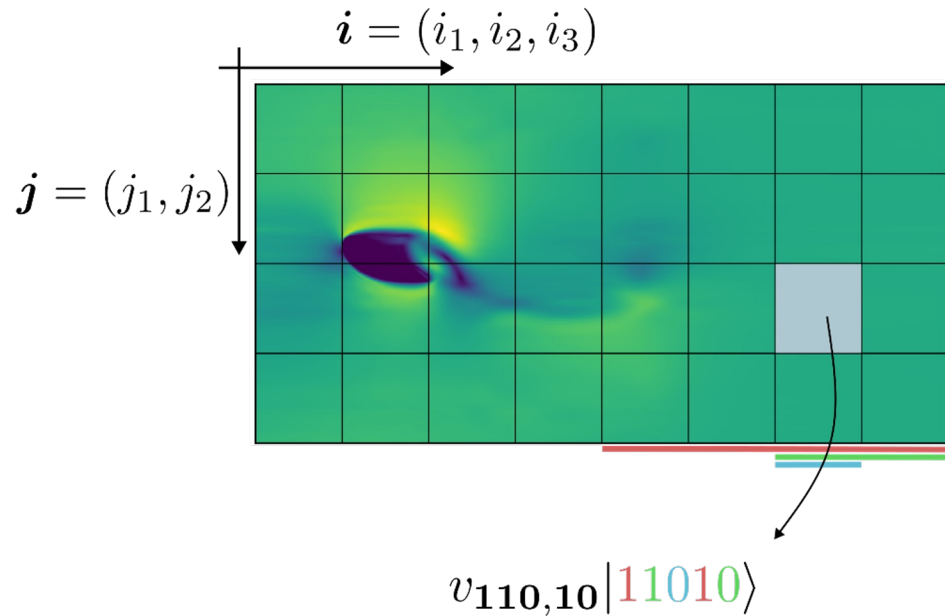
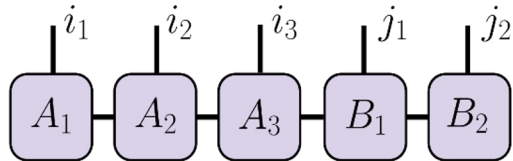
CFD framework

First step: Flow initialization
QTT encoding recap

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$$v_{\mathbf{i}, \mathbf{j}} = A_1^{(i_1)} A_2^{(i_2)} A_3^{(i_3)} B_1^{(j_1)} B_2^{(j_2)}$$



$$\mathcal{O}(2^N)$$



$$\mathcal{O}(N\chi^2)$$

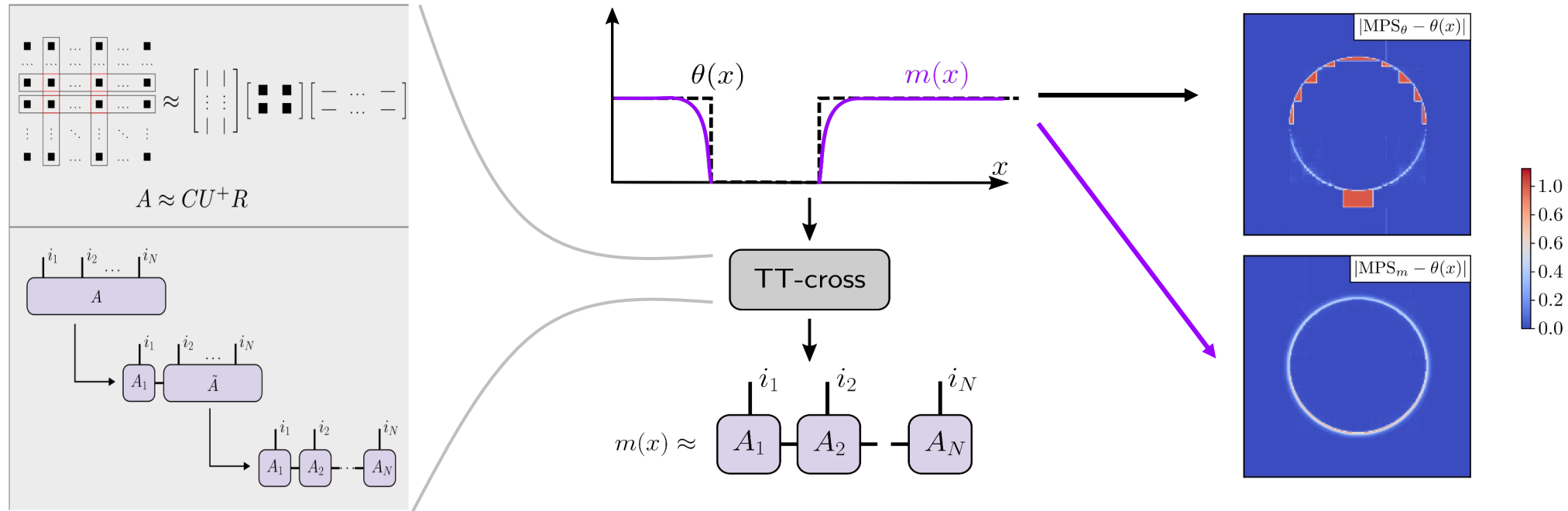
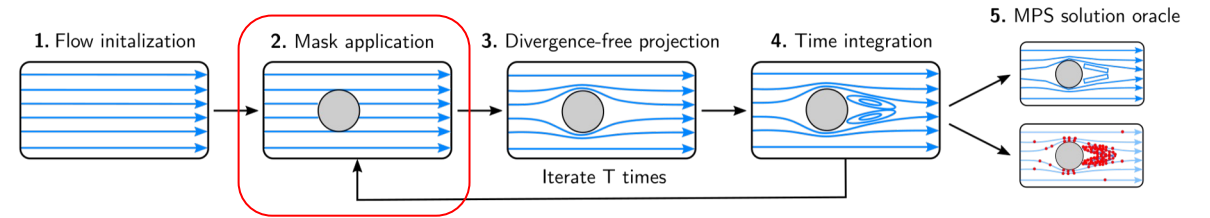
qubits $\Leftrightarrow$ scales
entanglement $\Leftrightarrow$ inter-scale correlations

CFD framework

Masking arbitrary objects

$$\mathcal{O}(N\chi^2)(\text{Cost. of eval.}) + \mathcal{O}(N\chi^3)$$

I. Oseledets and E. Tyrtshnikov, Linear Algebra and its Applications 432, 70 (2010)

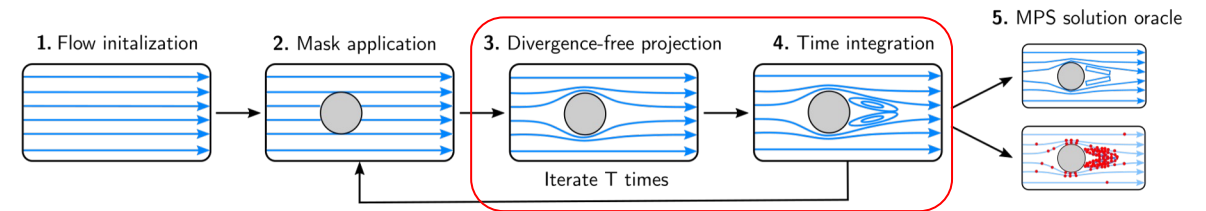


CFD framework

Time integrator

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v},$$

$$\nabla \cdot \mathbf{v} = 0$$



- Finite difference scheme
- MPO encoding of open boundary conditions (OBC)
- Chorin's projection method - predict and correct
- DMRG-based divergence-free projection - (lin. system) $\mathcal{O}(N\chi^6)$

Chorin's projection

$$\mathbf{v}_{t+\Delta t}^* = \mathbf{v}_t + (-(\mathbf{v}_t \cdot \nabla) \mathbf{v}_t + \nu \nabla^2 \mathbf{v}_t) \times \Delta t$$

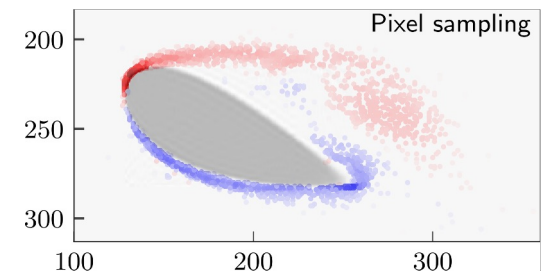
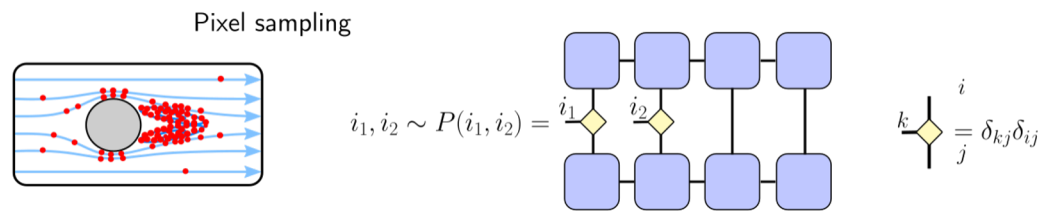
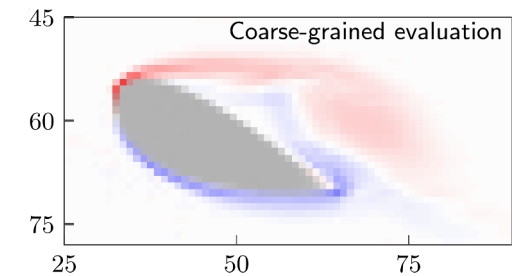
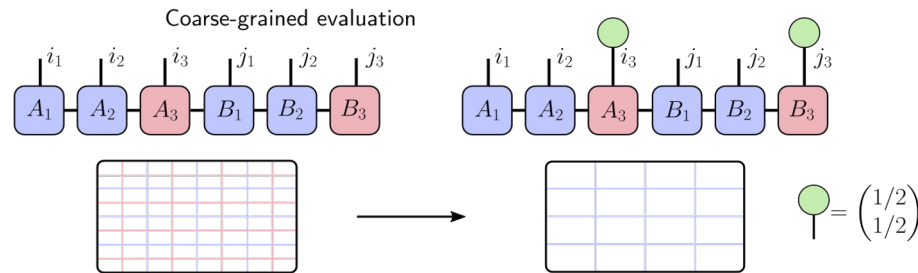
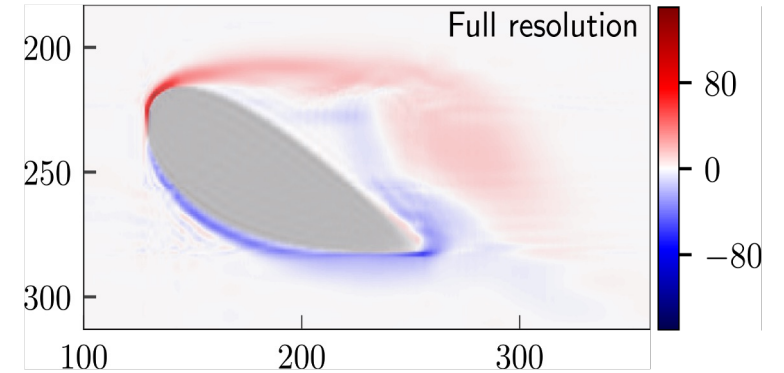
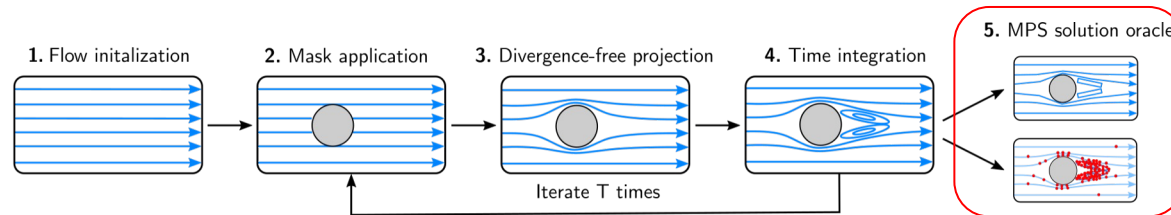
$$\nabla^2 p_{t+\Delta t} = \frac{\rho}{\Delta t} \nabla \cdot \mathbf{v}_{t+\Delta t}^*$$

$$\mathbf{v}_{t+\Delta t} = \mathbf{v}_{t+\Delta t}^* - \frac{\Delta t}{\rho} \nabla p_{t+\Delta t}$$

Integrate equations within the QTT representation

CFD framework

Recover the solution



$\mathcal{O}(N\chi^3)$

$\mathcal{O}(N_{\text{samples}}\chi^3)$

CFD framework

Complexity of the algorithms

Algorithmic task	Complexity	Runtime %
Mask generation	$\mathcal{O}(N \chi^3)$	offline
Mask application	$\mathcal{O}(\chi^6)$	2.0
Divergence-free projection	$\mathcal{O}(N \chi^6)$	80.1
Euler time stepping	$\mathcal{O}(\chi^6)$	17.9
Coarse-grained evaluation	$\mathcal{O}(N \chi^3)$	offline
Pixel sampling (per pixel)	$\mathcal{O}(\chi^3)$	offline
Numerically observed total runtime $\approx N \chi^{4.1}$		

← Approximate solutions possible $\sim \mathcal{O}(\chi^3)$

Computational Fluid Dynamics with QTT

Why fluid dynamics

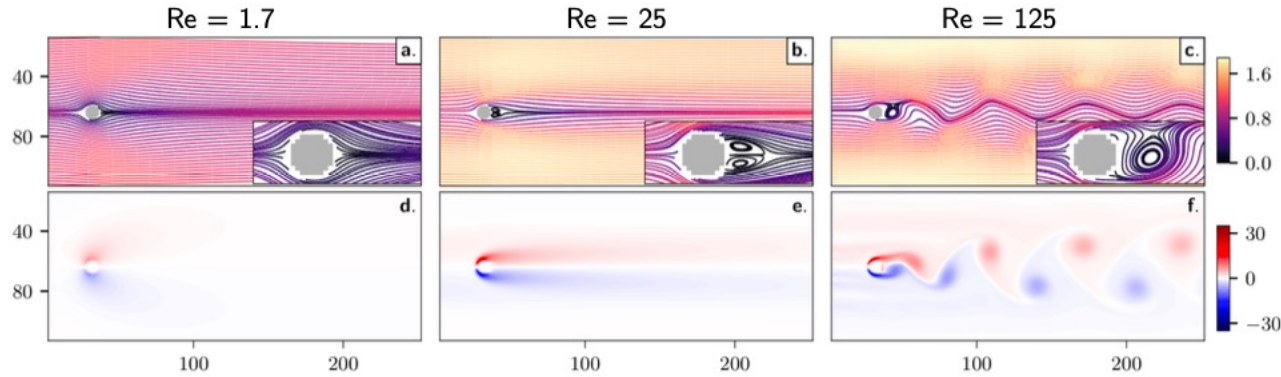
CFD framework

Numerical results

Recent literature

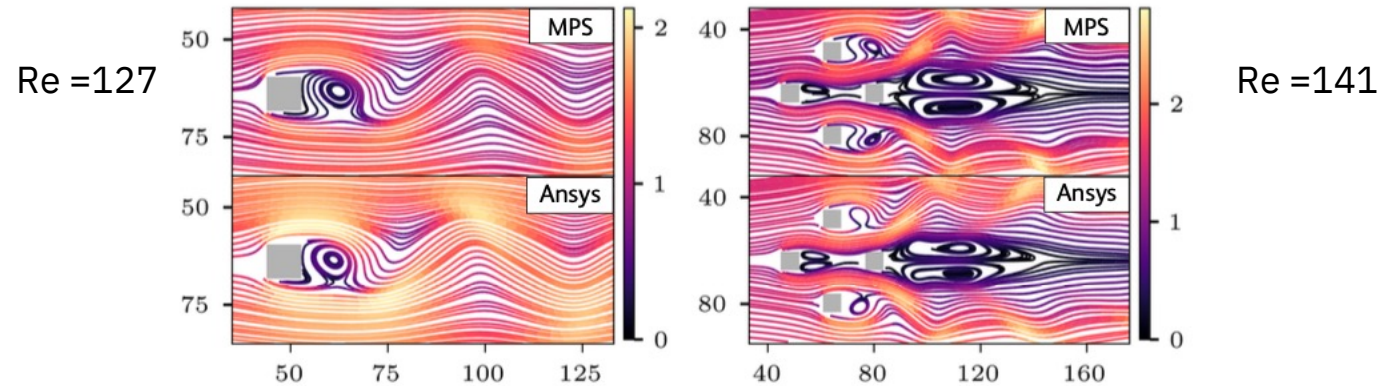
Toward turbulence

Numerical results



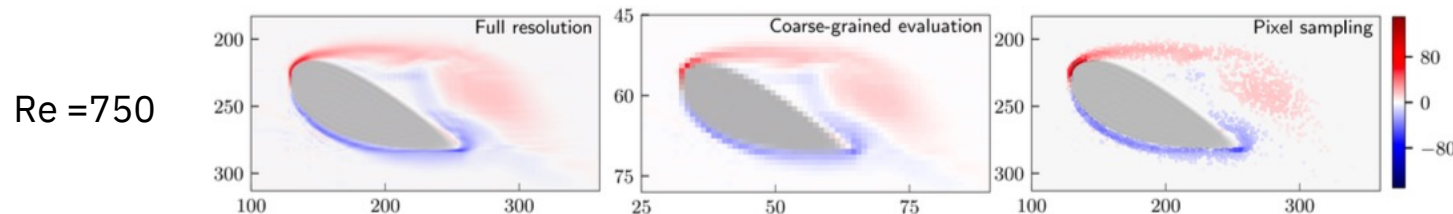
Flow around a cylinder

- Flow behaviour within the laminar regime
- 15-bit MPS encoding with $\chi = 30$
- Reproduced the Karman Vortex Street phenomena



Flow around squares

- MPS encoding of a collection of squares
- 15-bit MPS encoding with $\chi = 30$
- Qualitative comparison with Ansys
Fluent - a commercial CFD solver

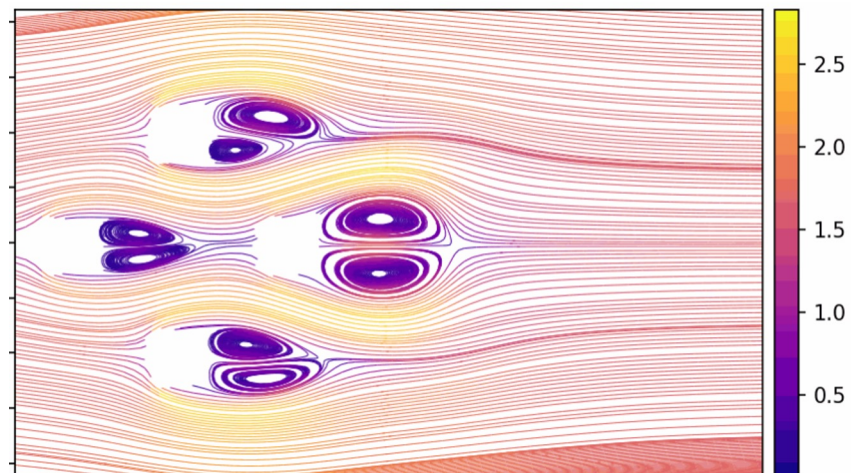
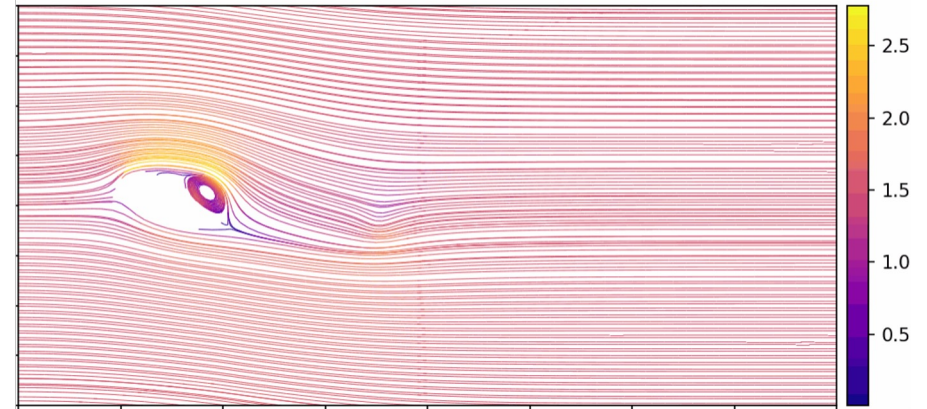
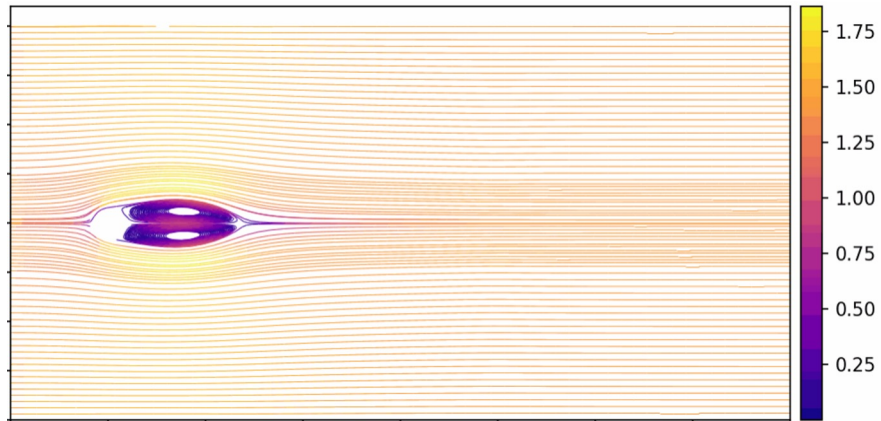


Flow around NACA airfoil

- MPS encoding of NACA 0040 inclined at 22°
- 19-bit MPS encoding with $\chi = 45$
- Various modes of the MPS solution oracle encoding the vorticity field

T=5000, 28x compression

Numerical results



$(100 < \text{Re} < 200)$

Computational Fluid Dynamics with QTT

Why fluid dynamics

CFD framework

Numerical results

Recent literature

Toward turbulence

Recent literature

Since the framework is flexible...

Technical report on a quantum-inspired solver for simulating compressible flows RD Peddinti, S Pisoni, E Tiunov, A Marini, L Aolita arXiv preprint arXiv:2506.03833	2	2025
Numerical solution of the incompressible Navier-Stokes equations for chemical mixers via quantum-inspired tensor train finite element method E Kornev, S Dolgov, K Pinto, M Pflitsch, M Perelshtein, A Melnikov arXiv preprint arXiv:2305.10784	31	2023
Matrix product state simulation of reacting shear flows R Pinkston, N Gourianov, H Alipanah, P Givi, D Jaksch, ... arXiv preprint arXiv:2512.13661	7	2025
Tensor network reduced order models for wall-bounded flows M Kiffner, D Jaksch Physical Review Fluids 8 (12), 124101	55	2023
Quantum-Inspired Simulation of 2D Turbulent Rayleigh-Bénard Convection NL van Hülst, MG Cecile, HY Van, T Hashizume, E de Villiers, D Jaksch arXiv preprint arXiv:2604.16179	1	2026
Quantum-inspired tensor-network fractional-step method for incompressible flow in curvilinear coordinates NL Van Hülst, P Siegl, P Over, S Bengoechea, T Hashizume, MG Cecile, ... Computer Physics Communications, 110169	14	2026

Recent literature

Since the framework is flexible...

Quantized tensor networks for solving the Vlasov–Maxwell equations E Ye, NF Loureiro Journal of Plasma Physics 90 (3), 805900301	38	2024
Quantum algorithm for the linear Vlasov equation with collisions A Ameri, E Ye, P Cappellaro, H Krovi, NF Loureiro 2023 IEEE International Conference on Quantum Computing and Engineering (QCE ...	31	2023
A Tensor-Train Discontinuous Galerkin Method for the Vlasov-Maxwell System R Chinomona, D Adak, WJ Barham, DP Truong, NV Roberts, ... arXiv preprint arXiv:2607.08936	1	2026
A tensor train-based isogeometric solver for large-scale 3D poisson problems QT Tran, DP Truong, KØ Rasmussen, B Alexandrov Computer Methods in Applied Mechanics and Engineering 453, 118802	6 *	2026
Tensor Train Neutron Transport Applied to Nuclear Reactor Problems MI Ortega, DP Truong, I Boureima, KØ Rasmussen, BS Alexandrov Nuclear Science and Engineering, 1-21	7	2026
Magic of the Well: assessing quantum resources of fluid dynamics data AF Mello, M Collura, EM Stoudenmire, R Levy arXiv preprint arXiv:2512.03177	1	2025
Full grid solution for multi-asset options pricing with tensor networks L Arenstein, M Kastoryano arXiv preprint arXiv:2601.00009	3	2025

...

Yes, but

We started from Turbulence

It is in the turbulent regime that we expect a memory advantage due to scale locality

The Navier-Stokes equation are dissipative → not norm preserving

We need to introduce the mathematical tools to benchmark and characterize turbulence

STATISTICAL METRICS

Note:

We'll still be using standard methods to compress the QTT (based on L2 norm)

What about other more clever constructions tailored to the problem?

Computational Fluid Dynamics with QTT

Why fluid dynamics

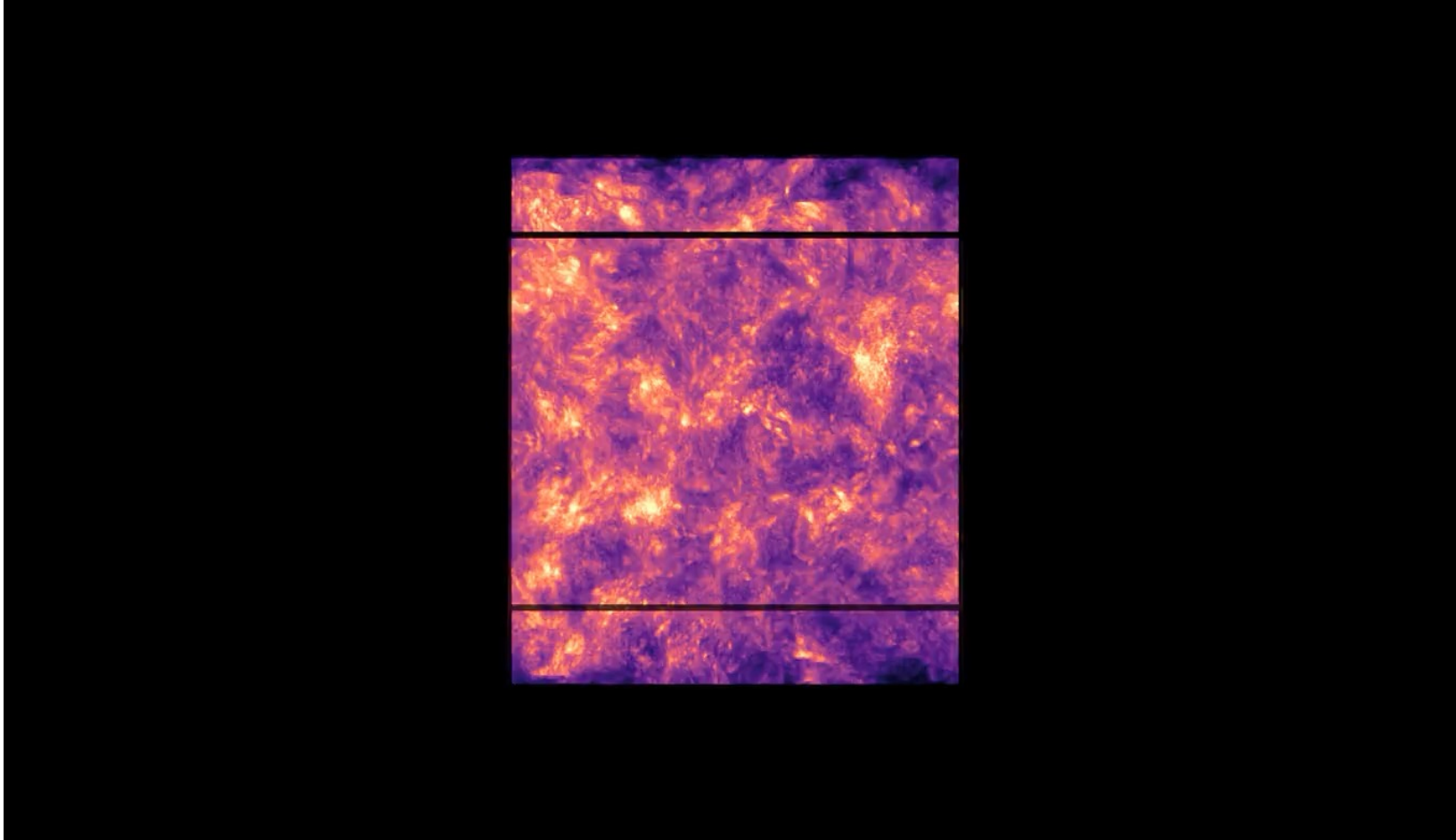
CFD framework

Numerical results

Recent literature

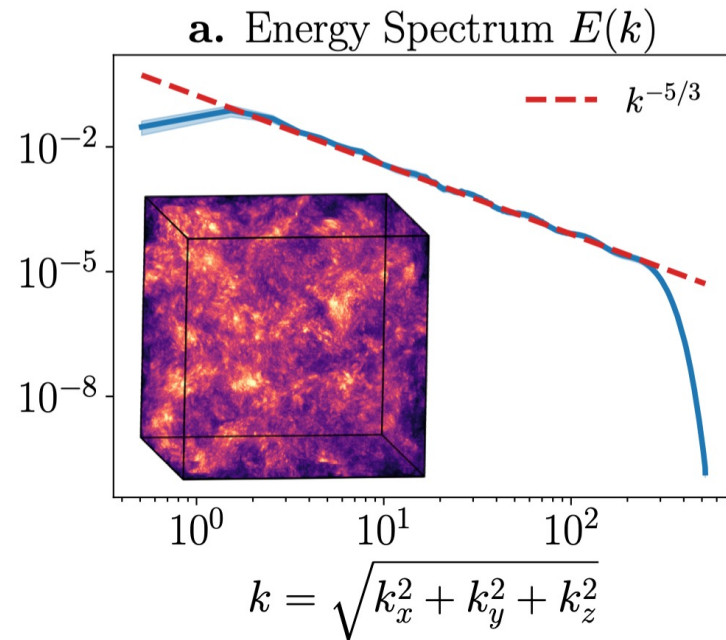
Toward turbulence

Snaphot generated natively in QTT



Toward turbulence

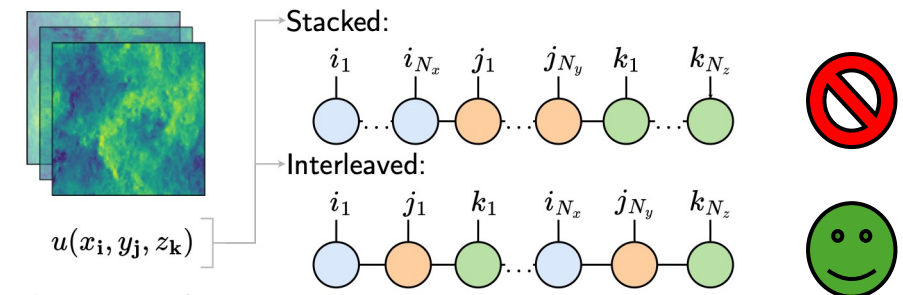
**3D DNS dataset: forced, isotropic,
periodic cube (1024^3), $Re_\lambda = 315$**
(Barcelona minotauro supercomputer, 2017)



PHYSICAL REVIEW RESEARCH 8, 023245 (2026)
***“Compression, simulation, and synthesis of
turbulent flows with tensor trains”***

Homogeneous Isotropic turbulence
↓
Compression, Evolution, Synthesis with QTT
↓
Statistical metrics:
 $E(k)$, Flatness, dv/dx PDF

Two QTT ordering: Stacked vs Interleaved



Toward turbulence

Statistical metrics

Kolmogorov (1941)

Scale-invariance $\rightarrow S_p(r) = \langle \delta v(r)^p \rangle \sim \epsilon^{p/3} r^{p/3}.$

$$E(k) \propto \epsilon^{2/3} k^{-5/3},$$

Deviations from scale inv. $\rightarrow$ intermittency

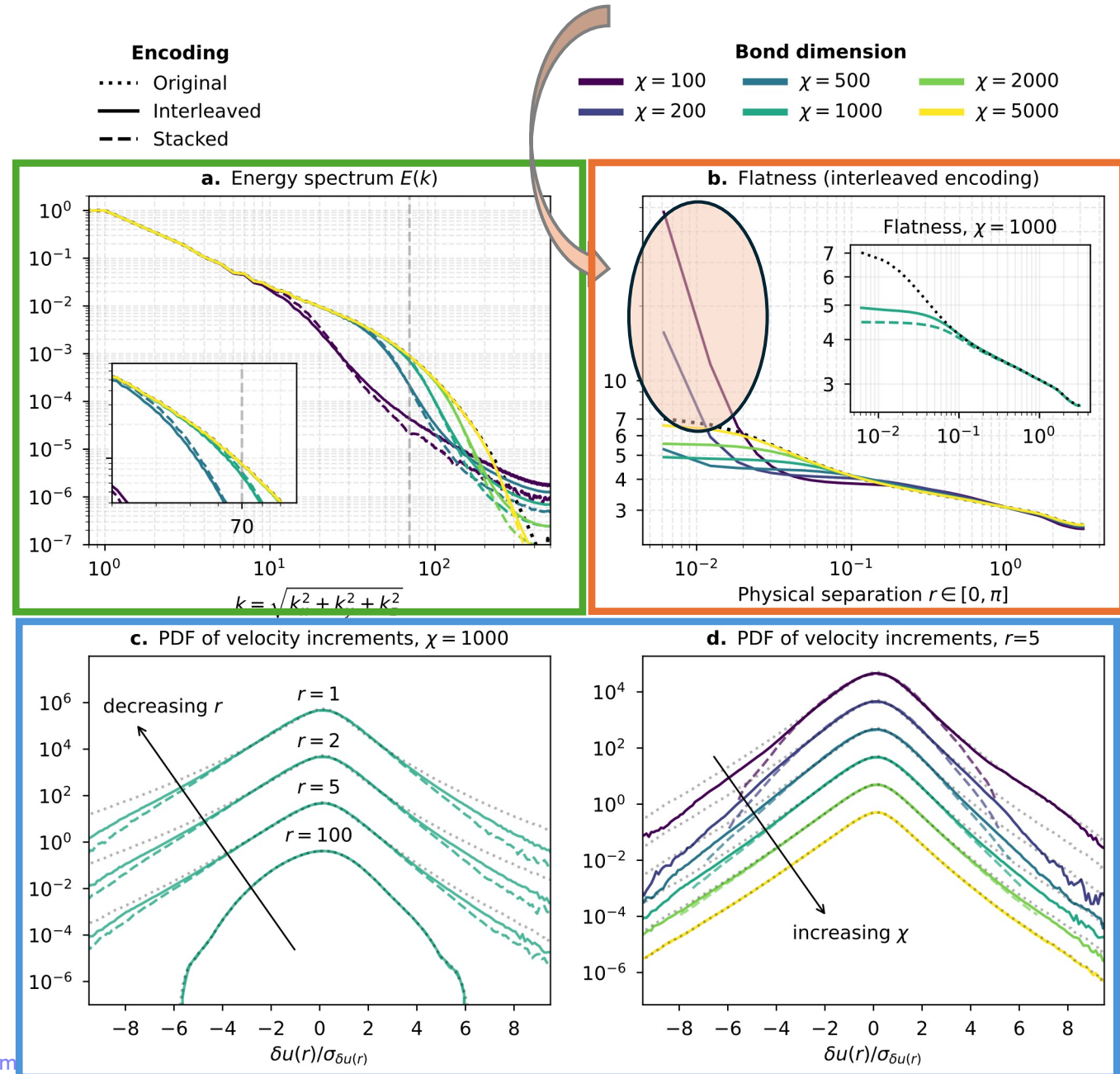
NON const. Flatness

Peaked PDF of vel. incr.

$$\Gamma_p(r) \equiv S_p(r)/S_2(r)^{p/2}.$$

$$\delta u(r)/\sigma_{\delta u(r)}$$

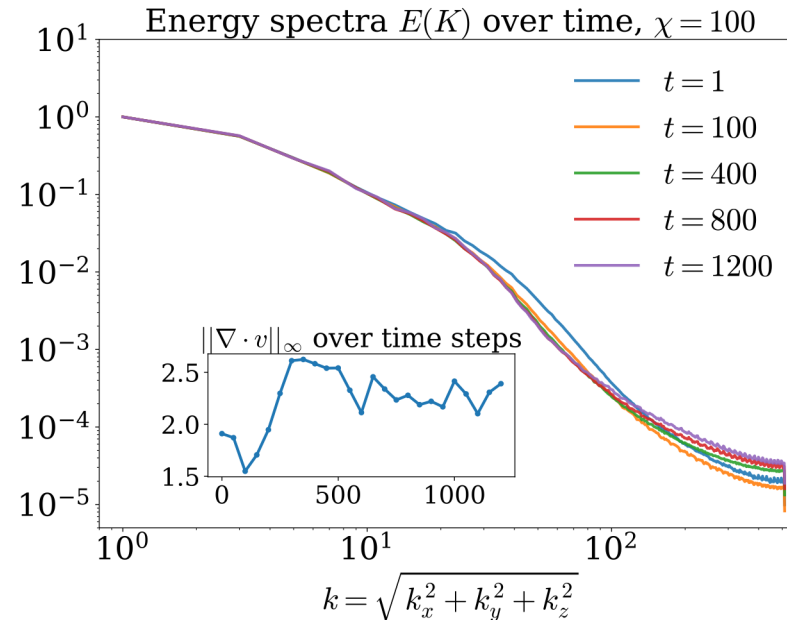
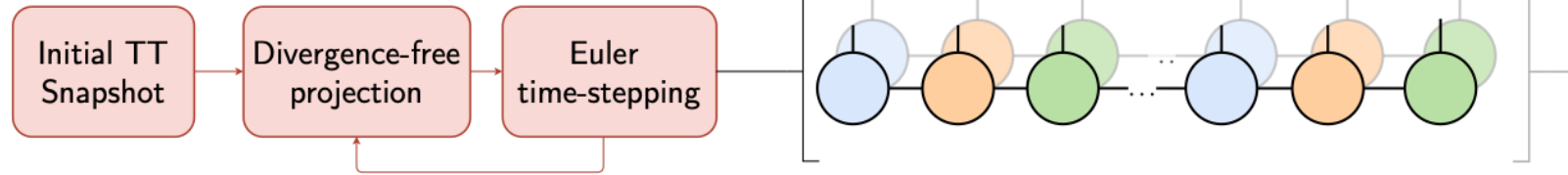
Good compression, but: Issue with small-scale physics!



Toward turbulence

Evolution scheme

B. 3D TT-based solver



9-10 Kolmogorov
time unit

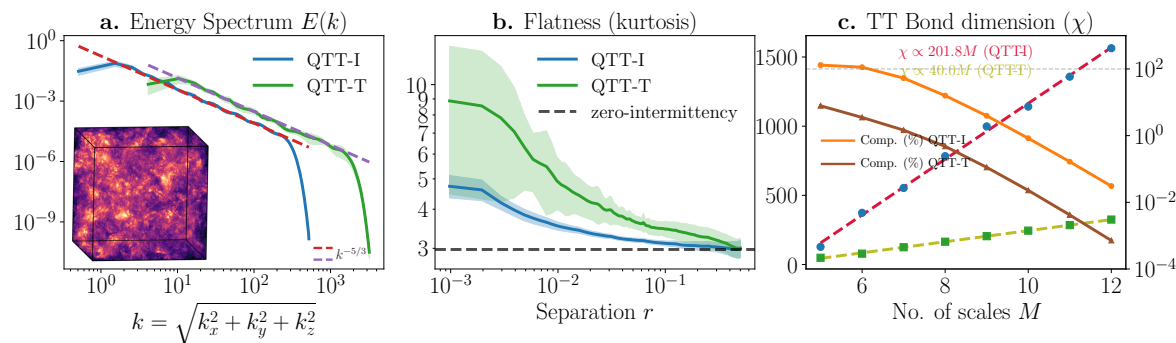
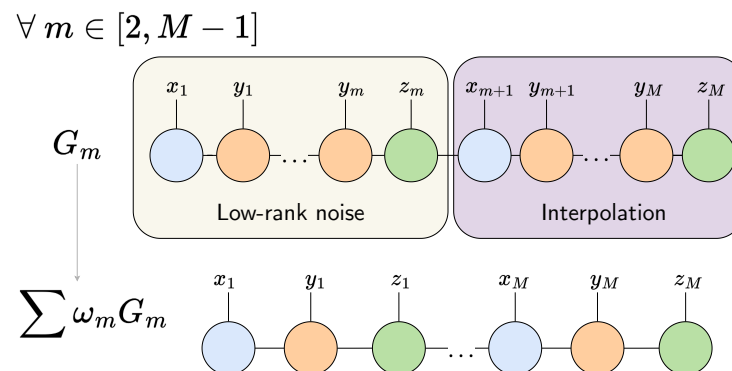
Toward turbulence

Generation scheme - interpolation

Efficient upsampling for tensor-network and quantum-state encoded functions

SE Guzman, E Tiunov, L Aolita
arXiv preprint arXiv:2601.03885

2 * 2026



Toward turbulence

Open questions

- Truncation VS Projection (3 vectorial components)
- Small scale overshooting for strong compressions

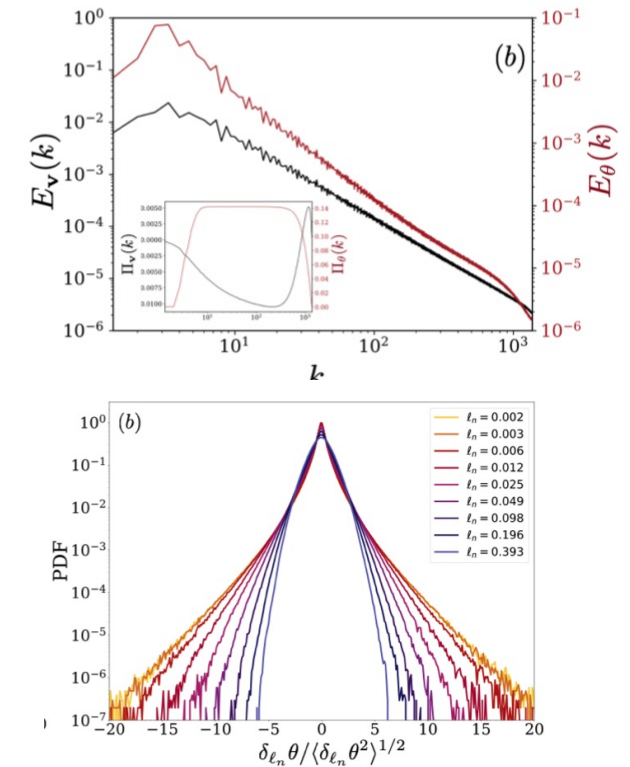
Passive scalar dataset

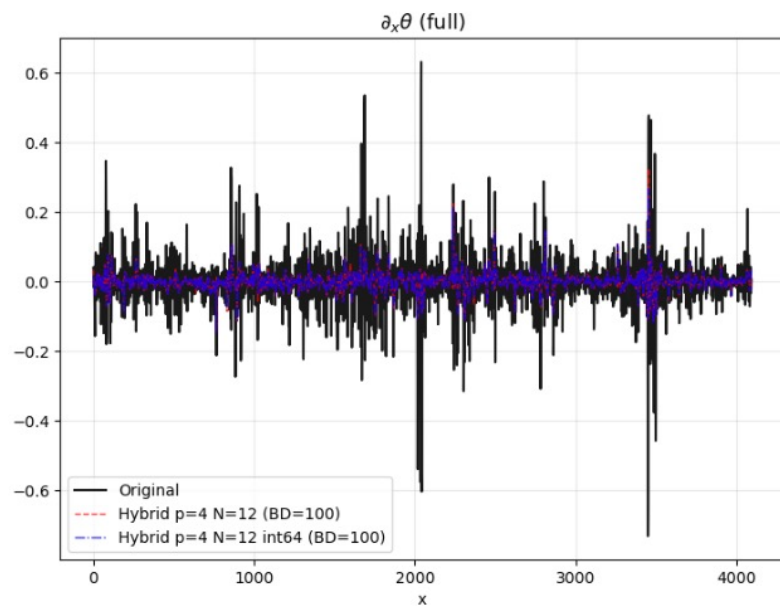
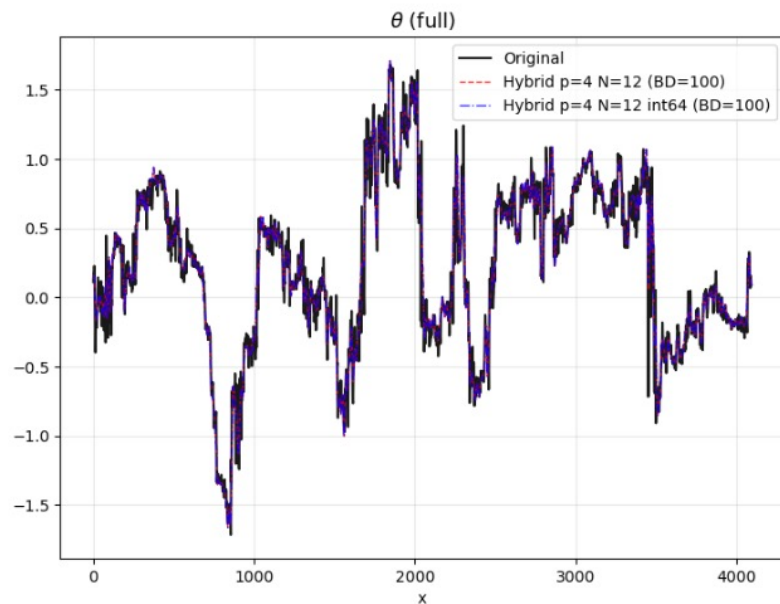
$$\partial_t \theta + \mathbf{v} \cdot \nabla \theta = \kappa \Delta \theta + f_\theta, \quad (1)$$

where κ is the scalar diffusivity, and f_θ is a large-scale Gaussian forcing [5]. The advecting velocity field $\mathbf{v}$ is incompressible and obeys the 2-dimensional Navier-Stokes equations:

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla p + \nu \Delta \mathbf{v} - \beta \mathbf{v} + \mathbf{f}, \quad (2)$$

where p is the pressure, ν the kinematic viscosity, β a large-scale friction coefficient that prevents the accumulation of energy due to the inverse cascade, and $\mathbf{f}$ is a small-scale Gaussian forcing.



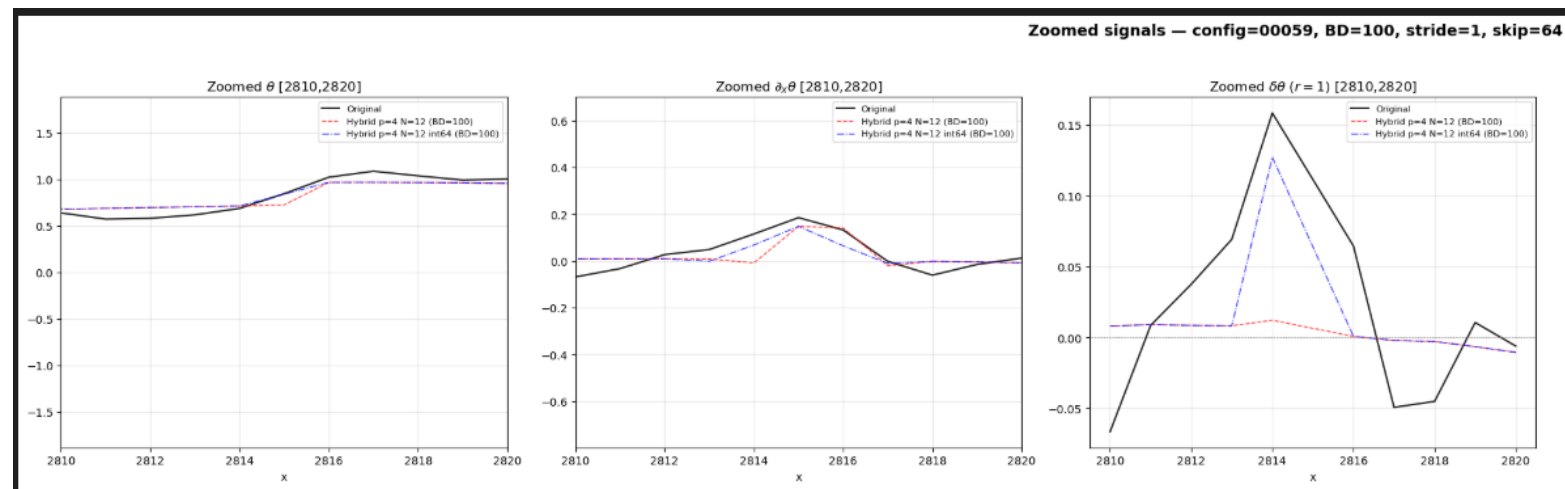


Toward turbulence

What is actually going on when we compress

Turbulence stats is about the standard deviation

The compression smoothens out the field, but at the block boundaries

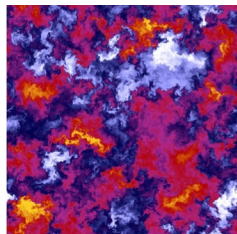


Toward turbulence

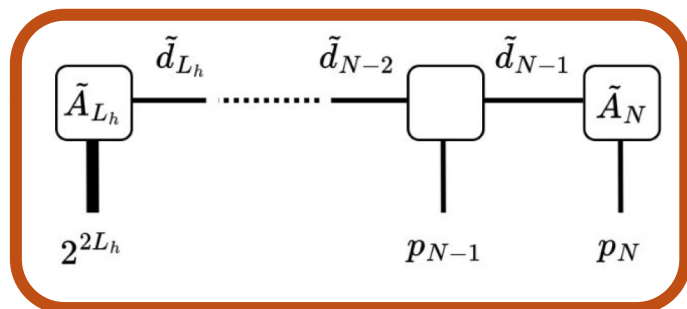
Solution to small scale unphysical behavior

2D passive scalar dataset, 4096x4096

(simulations from university of Rome II, Tor Vergata)



Compression via Hybrid-TT



arXiv:2608.00194v1 (2026)

**“Multiscale passive scalar turbulence
in a compressed subspace via tensor trains”**

Features:

Dense head tensor to retain large scales exactly
+
native QTT interpolation at the block boundaries

Results

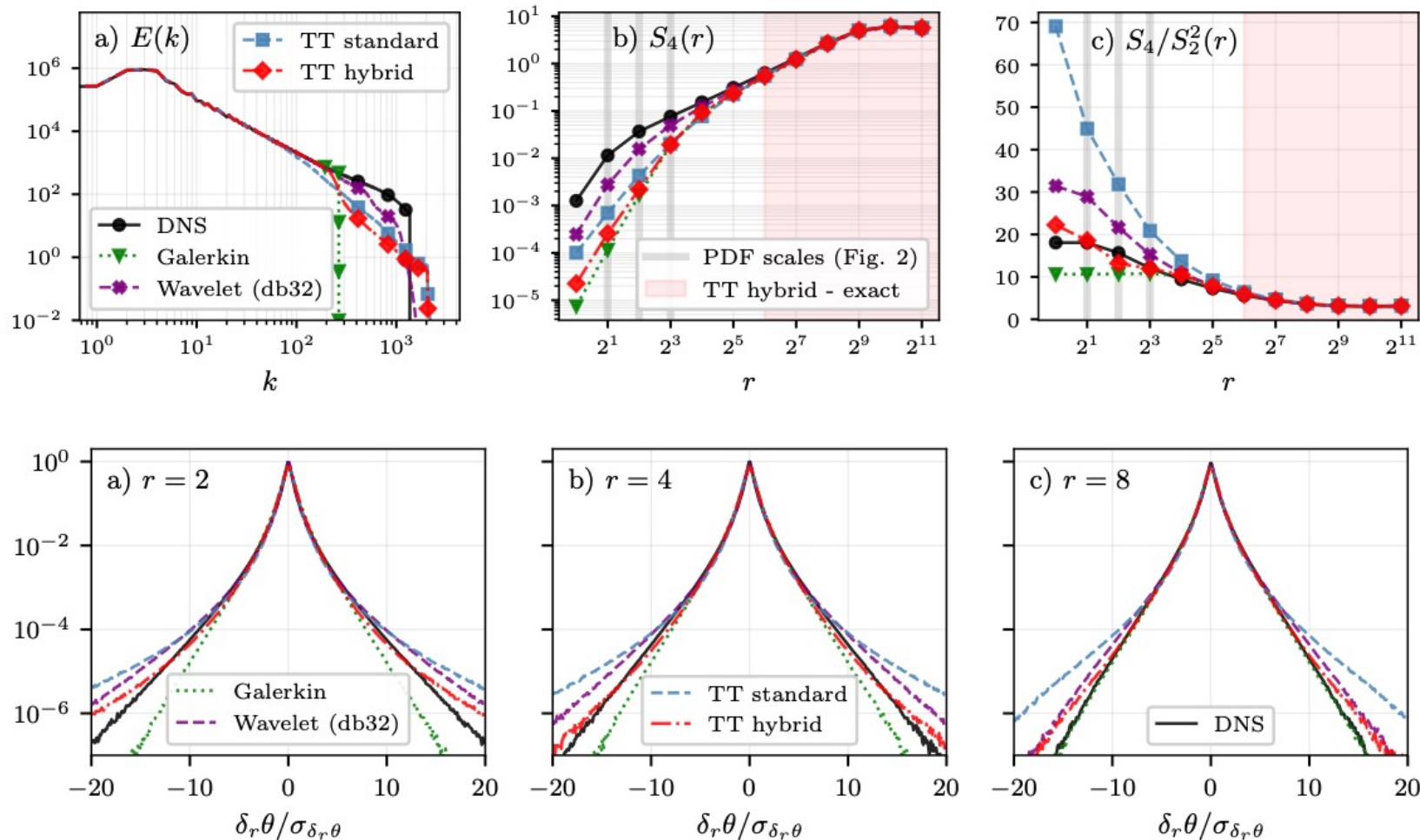
- Small scale statistics is well reproduced (fixed)!
- Better than Fourier and wavelet
- Study of the dependence on the head dimension and compression ratios

Toward turbulence

arXiv:2608.00194v1 (2026)

**“Multiscale passive scalar turbulence
in a compressed subspace via tensor trains”**

1.3% compression



Results

- Correct small scale statistics!
- Better than Fourier and wavelet!

More papers on turbulence

Quantum-inspired fluid simulation of two-dimensional turbulence with GPU acceleration

L Hölscher, P Rao, L Müller, J Klepsch, A Luckow, T Stollenwerk, ...
Physical Review Research 7 (1), 013112

42

2025

Tensor networks enable the calculation of turbulence probability distributions

N Gourianov, P Givi, D Jaksch, SB Pope
Science Advances 11 (5), eads5990

67

2025

A priori Assessment of Tensor-Network Encoding for Isotropic Turbulent Flows

M Esmaeili, H Alipanah, R Pinkston, P Givi, D Livescu, AJ Daley, ...
arXiv preprint, arXiv:2608.28869

2026